Adidas US Sales Performance Dashboard

2020 – 2021 | Power BI Desktop | Beginner Level

$899.9M Revenue · $332.1M Profit · 9,648 Records · 50 US States

TOOL	SOURCE	LEVEL	DATE RANGE	RECORDS
Power BI Desktop	Kaggle (Public)	Beginner	Jan 2020–Dec 2021	9,648 · 13 Cols

01 EXECUTIVE SUMMARY

- Processed and cleaned 9,648 sales records spanning 24 months (Jan 2020–Dec 2021) across 6 retailers, 5 US regions, 50 states, 52 cities, 6 product categories, and 3 sales channels — achieving 100% data integrity with zero null values across all 13 columns after Power Query ETL pipeline execution.
- Engineered 4 DAX measures (Total Revenue, Total Profit, Total Units, Avg Margin) and built a 12-visual interactive Power BI dashboard featuring KPI cards, clustered bar charts, donut chart, line chart, filled map, and 3 dynamic slicers enabling real-time cross-filtering across all dimensions.
- Analyzed $899,902,125 total revenue and $332,134,761 operating profit (42.30% avg margin), surfacing West Gear as top retailer ($242.96M), Men's Street Footwear as highest-revenue product ($208.83M), and West Region as dominant geography ($269.94M — 30.0% of total US revenue).
- Identified 294.23% revenue surge from 2020 ($182.08M) to 2021 ($717.82M), flagged Online channel as highest-margin sales method (38.98%), and pinpointed South Region as highest-margin geography (42.26%) — all surfaced through interactive dashboard filtering.

02 PROBLEM STATEMENT

Adidas US sales data spanning 2020–2021 existed as a flat Excel file with 9,648 records and no structured reporting layer. Business stakeholders had no visibility into which retailers, regions, or product categories were driving revenue and profit — nor was there any mechanism to compare year-on-year performance, identify underperforming geographies, or evaluate the relative profitability of In-store vs Outlet vs Online channels. Decisions were being made without data-backed insights, leaving significant performance gaps unaddressed across a $899.9M revenue portfolio.

03 OBJECTIVES

- **Obj 1:** Transform raw Excel data into a clean, analysis-ready dataset through structured ETL steps in Power BI Power Query.

- **Obj 2:** Create calculated DAX measures enabling dynamic aggregation of revenue, profit, units sold, and operating margin.
- **Obj 3:** Build a 12-visual interactive dashboard covering retailer performance, regional breakdown, product mix, monthly trends, and geographic distribution across 50 US states.
- **Obj 4:** Enable drill-down analysis via 3 dynamic slicers (Region, Retailer, Sales Method) allowing instant cross-filtering across all dashboard visuals simultaneously.
- **Obj 5:** Surface actionable insights identifying top-performing and underperforming segments across all 4 business dimensions.
- **Obj 6:** Produce a portfolio-ready Power BI report with 3 custom JSON themes demonstrating end-to-end data preparation, modelling, and visualisation skills.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Adidas US Sales Dataset
Source	Kaggle — heemalichaudhari/adidas-sales-dataset
File Format	Microsoft Excel Workbook (.xlsx)
Total Records	9,648 rows (after removing 4 junk header rows)
Total Columns	13 usable columns (after removing blank index column)
Date Coverage	January 1, 2020 – December 31, 2021 (24 months)
Geographic Scope	50 US states · 52 unique cities · 5 geographic regions
Null Values	Zero — 100% complete, no imputation required
Header Issue	Title 'Adidas Sales Database' at row 2; actual headers at row 5 — required ETL fix

Column Definitions (13 Columns)

Column	Type	Description
Retailer	Text	6 retail partners (West Gear, Foot Locker, Sports Direct, Kohl's, Amazon, Walmart)
Retailer ID	Whole Number	Unique numeric ID per retailer
Invoice Date	Date	Transaction date — daily granularity, Jan 2020–Dec 2021
Region	Text	5 US regions: Northeast, South, West, Midwest, Southeast
State	Text	50 unique US states
City	Text	52 unique cities
Product	Text	6 categories: Men's/Women's Street Footwear, Athletic Footwear, Apparel
Price per Unit	Decimal	Unit selling price in USD (average: $45.22)
Units Sold	Whole Number	Quantity sold per transaction line

Total Sales	Decimal	Revenue = Price per Unit × Units Sold
Operating Profit	Decimal	Profit after operating costs per transaction
Operating Margin	Percentage	Profit margin per line (average: 42.30%)
Sales Method	Text	3 channels: In-store ($356.6M), Outlet ($295.6M), Online ($247.7M)

05 TOOLS & TECHNOLOGIES USED

Tool / Technology	Platform	Purpose in This Project
Power BI Desktop	Microsoft (Latest 2024)	Primary BI tool for ETL, modelling, DAX creation, and dashboard visualisation
Power Query Editor	Built into Power BI	ETL pipeline — data cleaning, type conversion, header promotion, column removal
DAX (Data Analysis Expressions)	Power BI Native Language	4 calculated measures: Total Revenue, Total Profit, Total Units, Avg Margin
Microsoft Excel (.xlsx)	Source File Format	Raw dataset stored in Excel — imported via Get Data → Excel Workbook connector
Kaggle	Public Dataset Platform	Free public dataset source — no API key required, CSV/Excel download
JSON Theme Files	Power BI Custom Theme	3 portfolio themes: Arctic Pro (Light), Command Center (Dark), Fusion Indigo (Mix)

06 ETL PROCESS — DATA PREPARATION IN POWER QUERY

Raw Excel file contained a 4-row header section (title row + blank separator rows) above actual data headers, an unnamed empty first column, and 6 columns with incorrect data types requiring correction. All 10 transformation steps were executed in Power BI Power Query Editor — creating a repeatable, auto-refreshable ETL pipeline logged in the Applied Steps pane.

Step	Power Query Action	Reason / Impact
01	Get Data → Excel Workbook → selected 'Data Sales Adidas' sheet	Established live source connection to raw .xlsx file
02	Clicked 'Transform Data' (NOT Load) → opened Power Query Editor	Enabled pre-load cleaning; prevents dirty data entering the data model
03	Home → Remove Rows → Remove Top Rows → entered 4	Removed 'Adidas Sales Database' title and 3 blank separator rows
04	Home → Use First Row as Headers	Promoted correct column names (Retailer, Invoice Date…) from row 5
05	Right-clicked Column1 (blank/null) → Remove	Eliminated empty unnamed index column inherited from Excel formatting
06	Invoice Date column type → Date	Enabled time-intelligence, date hierarchy, monthly trend line chart
07	Operating Margin column type → Percentage	0.42 decimal now displays as 42% correctly in all visuals
08	Price per Unit, Total Sales, Operating Profit → Decimal Number	Enabled correct SUM/AVERAGE currency aggregations
09	Units Sold column type → Whole Number	Correct integer type for quantity fields; prevents decimal unit display
10	Home → Close & Apply	Loaded 9,648 clean rows into Power BI model; 0 errors, 0 nulls, 0 data quality issues

07 DAX MEASURES CREATED

4 DAX measures were created via right-click on table → New Measure. Measure names were intentionally differentiated from existing column names to prevent Power BI naming conflict errors (e.g., 'Total Revenue' instead of 'Total Sales' which already existed as a column).

Measure Name	DAX Formula	Result / Use
Total Revenue	`SUM('Data Sales Adidas'[Total Sales])`	$899,902,125 — primary KPI; used in all bar charts and map
Total Profit	`SUM('Data Sales Adidas'[Operating Profit])`	$332,134,761 — profitability KPI card

Total Units	`SUM('Data Sales Adidas'[Units Sold])`	2,478,861 units — volume analysis KPI card
Avg Margin	`AVERAGE('Data Sales Adidas'[Operating Margin])`	42.30% — efficiency metric; responds to all slicer filters

08 DASHBOARD VISUALS BUILT (12 Visuals)

#	Visual Type	Fields Used	Insight Delivered
1	KPI Card	Total Revenue measure	$899.9M total US Adidas revenue 2020–2021
2	KPI Card	Total Profit measure	$332.13M operating profit portfolio-wide
3	KPI Card	Total Units measure	2,478,861 total units sold across all channels
4	KPI Card	Avg Margin measure	42.30% average operating margin
5	Clustered Bar Chart	Y: Retailer / X: Total Revenue	West Gear #1 at $242.96M (27% revenue share)
6	Clustered Bar Chart	Y: Region / X: Total Revenue	West leads at $269.94M (30% of US total)
7	Donut Chart	Legend: Product / Values: Total Revenue	Men's Street Footwear top at $208.83M (23.2%)
8	Line Chart	X: Invoice Date / Y: Total Revenue	294.23% YoY growth spike 2020→2021 visualised
9	Filled Map	Location: State / Color: Total Revenue	NY, CA, FL top 3 revenue-generating states
10	Slicer	Region field (5 values)	Filters all 9 visuals by US geographic region
11	Slicer	Retailer field (6 values)	Filters all visuals by retail partner
12	Slicer	Sales Method field (3 values)	Filters by In-store / Outlet / Online channel

09 KEY INSIGHTS & RESULTS

A. Revenue & Profitability Overview

- Analyzed $899,902,125 total US Adidas revenue and $332,134,761 operating profit across 9,648 transactions — delivering 42.30% average operating margin with $45.22 average unit price across 2,478,861 total units sold.
- Identified 294.23% YoY revenue growth from $182,080,675 (2020) to $717,821,450 (2021), with operating margin improving from 34.81% to 37.44% — signalling accelerated post-pandemic demand recovery with improving cost efficiency.

B. Retailer Performance

- West Gear ranked #1 retailer at $242,964,333 revenue (27.0% share) and $85,667,873 profit; Foot Locker followed at $220,094,720; Sports Direct delivered highest operating margin at 40.74% despite ranking 3rd in revenue at $182,470,997.
- Walmart flagged as lowest-margin retailer at 34.58% on $74,558,410 revenue; Amazon achieved 37.09% margin on $77,698,912 revenue — with Online channel efficiency partially offsetting lower unit volumes (197,990 units).

C. Regional Breakdown

- West Region dominated at $269,943,182 (30.0% of total), followed by Northeast $186,324,067 (20.7%), Southeast $163,171,236 (18.1%), South $144,663,181 (16.1%), and Midwest $135,800,459 (15.1%).
- South Region achieved highest operating margin at 42.26% on $144.66M revenue — outperforming the national 42.30% avg and exceeding West Region's 33.20% margin by 9.06 percentage points, flagging profitable under-scaled geography.

D. Product Category Analysis

- Men's Street Footwear generated highest revenue at $208,826,244 (23.2% share) with 39.65% margin and 593,320 units — Adidas' clear flagship US category; Women's Apparel ranked 2nd at $179,038,860 with 38.34% margin.
- Men's Athletic Footwear ranked 3rd at $153,673,680 but carried lowest margin at 33.74% — 8.56pp below the portfolio average — indicating pricing pressure or higher cost structure in the athletic segment requiring strategic review.

E. Sales Channel Analysis

- Online channel delivered highest margin at 38.98% on $247,672,882 revenue and 939,093 units — highest unit volume across all channels. Despite 3rd-place revenue, superior margin profile identifies Online as highest-priority margin expansion channel.
- In-store led revenue at $356,643,750 (39.6% of total) but carried the lowest margin at 35.78%; Outlet generated $295,585,493 at 36.53% across 849,778 units — balanced mid-tier position with strong volume throughput.

F. Geographic Distribution

- New York ($64,229,039), California ($60,174,133), and Florida ($59,283,714) collectively contributed $183,686,886 — 20.4% of total US revenue across just 3 of 50 states, highlighting significant geographic revenue concentration.
- Nebraska ($5,929,038), Minnesota ($7,378,068), and Iowa ($7,424,011) ranked as lowest-revenue states — collectively representing just 2.3% of total US revenue — flagging untapped market penetration opportunity across the upper Midwest region.

KPI Summary

KPI Metric	Value	KPI Metric	Value
Total Revenue	**$899,902,125**	**YoY Growth 2020→2021**	294.23%
Total Operating Profit	**$332,134,761**	**2020 Revenue**	$182,080,675
Total Units Sold	**2,478,861**	**2021 Revenue**	$717,821,450
Avg Operating Margin	**42.30%**	**Avg Price per Unit**	$45.22
Top Retailer	**West Gear — $242.96M**	**Lowest Margin Retailer**	Walmart — 34.58%
Top Region	**West — $269.94M (30%)**	**Highest Margin Region**	South — 42.26%
Top Product	**Men's Street Footwear — $208.83M**	**Lowest Margin Product**	Men's Athletic — 33.74%
Highest Margin Channel	**Online — 38.98%**	**Highest Revenue Channel**	In-store — $356.64M
Top State	**New York — $64.23M**	**Lowest Revenue State**	Nebraska — $5.93M
Unique Retailers	**6**	**Unique Products**	6
US States Covered	**50**	**Unique Cities**	52

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Malformed Excel Header Structure

- **Problem:** Problem: Raw file had a 'Adidas Sales Database' title row and 3 blank rows above actual column headers, causing Power BI to import all columns as 'Column1, Column2...' with junk data in the first 4 rows of the dataset.
- **Solution:** Solution: Used Power Query — Home → Remove Rows → Remove Top Rows (4) → then Use First Row as Headers — precisely promoting actual column names. Verified all 13 column names appeared correctly in the column header bar before proceeding to the next step.

Challenge 2 — DAX Naming Conflict Error

- **Problem:** Problem: Attempting to create a DAX measure named 'Total Sales' triggered Power BI error: 'The name Total Sales is already used for a column on table Data Sales Adidas. Choose a different name.' — blocking measure creation entirely.
- **Solution:** Solution: Renamed the measure to 'Total Revenue' to differentiate from the source column. Applied same convention to all 4 measures — Total Revenue, Total Profit, Total Units, Avg Margin — ensuring zero naming conflicts and clear semantic distinction between source columns and calculated measures.

Challenge 3 — Operating Margin Displaying as Decimal

- **Problem:** Problem: Operating Margin values were stored as raw decimals (0.42, 0.35...) in the Excel source. Without correcting the data type in Power Query, Power BI displayed '0.42' in KPI cards and charts instead of the correct '42%' format.
- **Solution:** Solution: Changed Operating Margin column type to 'Percentage' in Power Query before loading. This caused the values to render correctly as 42.30% across all visuals, cards, and tooltips automatically without requiring any additional visual-level formatting.

Challenge 4 — Blank Index Column Pollution

- **Problem:** Problem: Excel file contained an unnamed empty column (inherited from Excel's row-number structure) that appeared as 'Column1' filled with null values in Power Query — polluting the column list and the data model's field pane.
- **Solution:** Solution: Right-clicked the blank column header → Remove in Power Query before closing and applying. Eliminated it at the source level so it never entered the data model, never appeared in the field pane, and never caused confusion when building visuals.

Challenge 5 — Manual Visual Formatting at Scale

- **Problem:** Problem: Formatting 12 individual visuals (background, border, title colour, axis labels, legend text) manually for a professional portfolio look would require 60+ individual formatting changes — extremely time-consuming and inconsistent.
- **Solution:** Solution: Authored 3 custom Power BI JSON theme files encoding all visual styling (dataColors, page background, card borders, slicer headers, axis label colours). Single import via View → Themes → Browse for Themes instantly re-themed all 12 visuals simultaneously, saving approximately 2 hours of manual formatting.

11 SKILLS LEARNED & TAKEAWAYS

- **Power Query ETL Pipeline:** Executed 10-step data cleaning sequence — Remove Top Rows, Use First Row as Headers, column removal, 6 data type corrections — building a fully repeatable, auto-refreshable transformation pipeline logged in the Applied Steps pane.

- **DAX Fundamentals:** Created 4 calculated measures using SUM() and AVERAGE() functions; understood the critical distinction between measures (dynamic, filter-context aware) and columns (static, row-level); resolved real naming conflicts with a production workaround.
- **Data Type Awareness:** Experienced first-hand how storing Operating Margin as Decimal vs Percentage directly impacts visual rendering — learning to fix data types at source (Power Query) rather than patching individual visuals downstream.
- **Visual Composition for Dashboards:** Designed a 12-visual layout balancing KPI summaries, comparison charts, trend analysis, geographic distribution, and interactive filters — developing judgement for which visual type best communicates each data pattern.
- **Cross-Filter Interactions:** Understood Power BI's default cross-filtering — clicking a bar in the retailer chart auto-filters the map, donut, and line chart simultaneously — and how slicers extend this to user-driven, multi-dimensional data exploration.
- **Custom JSON Theming:** Authored Power BI theme JSON files — learning schema structure (dataColors array, visualStyles per visual type, page background, title bg) — enabling professional portfolio-grade aesthetics without per-visual manual formatting.
- **Business Insight Extraction:** Translated 9,648 transactional records into executive-level findings — highest-margin channel (Online: 38.98%), lowest-margin retailer (Walmart: 34.58%), geographic concentration risk (top 3 states = 20.4% of revenue), and 294.23% YoY growth story.

12 CONCLUSION

- Processed 9,648 Adidas US sales records through a structured 10-step Power Query ETL pipeline — removing 4 junk header rows, correcting 6 column data types, eliminating 1 blank column — achieving 100% clean data load with zero nulls across all 13 fields into the Power BI data model.
- Engineered 4 DAX measures (Total Revenue, Total Profit, Total Units, Avg Margin) and built a 12-visual interactive dashboard spanning KPI cards, comparison bar charts, donut chart, monthly trend line, filled US state map, and 3 cross-filtering slicers — zero code written, entirely no-code Power BI workflow.
- Surfaced $899.9M in total revenue, $332.13M profit (42.30% avg margin), West Gear as top retailer ($242.96M), West Region as dominant geography ($269.94M — 30% US share), and Men's Street Footwear as flagship product ($208.83M — 23.2% revenue share).
- Identified Online as highest-margin channel (38.98%), South Region as margin leader (42.26%), Men's Athletic Footwear as lowest-margin product (33.74%), and a 294.23% YoY revenue growth spike from $182.08M (2020) to $717.82M (2021) — actionable insights across 4 business dimensions.
- Delivered 3 portfolio-grade custom JSON themes (Arctic Pro Light, Command Center Dark, Fusion Indigo Mix) demonstrating end-to-end command of Power BI from raw Excel ingestion through ETL, DAX modelling, visual design, and professional theming — Project 1 of 9 in the Data Analytics Portfolio.